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Predictive Maintenance in CNC Machines Using Machine Learning

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***Abstract*— With the Computer Numerical Control (CNC) machining, predictive maintenance has been highly instrumental in the elimination of service breakdowns, and overall cost of operation. The specified article suggests the machine learning technique to identify the early warning of the possible machine failure relying on the data of the spindle vibration, temperature, and sound signal as indicated by the sensors. In various load and wear conditions, we have learned a set in a 3-axis CNC mill machine and used some of the supervised learning methods, e.g. Random Forest, Support Vector Machine (SVM) and LSTM networks. Of all these, LSTM model has the best performance with 94.7% accuracy over SVM (88.2%) and Random Forest (91.3%) in the predictability of the maintenance states. More so, we minimized outages by 27% in comparison to planned maintenance schedules. The findings decide the efficiency of such combination of time-series analysis and so as to prevent the anticipated failed working of the CNC machines earlier. The brought system can be scaled and the dependency on the human capacity is minimum as well as the system is far more secure in terms of reliability and productivity.**

Keywords*— *Predictive Maintenance, CNC Machines, Machine Learning, LSTM, Condition Monitoring

I. Introduction

Predictive maintenance in Computer Numerical Control (CNC) machining has risen as an efficient measure of reducing the occurrence of unforeseen failure, increasing the machine lifespan, and streamlining the maintenance models. In contrast to preventive maintenance based on fixed intervals, data-driven methods employ multi-sensor feedbacks and machine learning to forecast faults, which is in line with Industry 4.0 objectives of intelligent manufacturing [1]. Although there has been significant improvement in the period between 2020 and 2025, significant challenges remain. Current research on digital twins and the learning-based diagnostics does not integrate sensor and simulation data into the real world in general [2]. Deep learning and risk-based models are more accurate but have impediments to their deployment because of the heterogeneity of data and computing needs [3]. Moreover, there is still a lack of model generalization among machines and the inability to trust the explanations of non-explainable architectures [4], [5], as well as a lack of responsiveness in real-time due to the overhead of processing [6].

This paper fills these gaps by proposing a predictive maintenance system (based on machine learning) that is more optimized to CNC settings. It also has multimodal sensor data (vibration, temperature, acoustic) and explainable AI to be interpreted by users and guarantees reliable and real-time functionality. It compares the Random Forest (RF), Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) models and assesses the predictive accuracy, interpretability and efficiency because it is going to contribute towards better maintenance schedules and operational standards.

The major contributions of this work are as follows:

1. An all-purpose hybrid predictive maintenance pipeline, which will be in a position to capture and process multimodal CNC sensor data, backed up by custom feature engineering and dimensionality reduction.
2. Comparison of the classical ML models (RF, SVM, XGBoost) with deep learning models (LSTM) and the use of explainability to determine the most significant features.
3. A real-time decision support prototype that can help produce actionable maintenance alerts which can minimize downtime by 27% in experimental validation.
4. A feature-importance analysis that can be interpreted to show that spectral entropy, vibration RMS, and temperature gradients are the important predictors of tool degradation.

This paper presents a single, unified predictive maintenance system of CNC machines that combines vibration, temperature, and acoustic data with explainable AI (XAI) to be deployed in real-time and on the edge. In comparison to single-sensor models or opaque models, it benchmarks SVM, RF, and LSTM on an equal basis of preprocessing and tests it in real-life industrial application, where the model is fair and practically viable. This implies a resultant product, interpretable system, that enhances predictive reliability, minimizes downtimes and suits many different CNC configurations.

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The paper has the following structure. Related work and gaps are reviewed by Section II, the experimental setup as well as data, preprocessing, feature and math are covered in Section III, and it also contains models, training, XAI and system architecture, Section IV reports the results (performance, feature importance and real-time evaluation) as well as the comparative study and Section V concludes with findings, limitations and future work.

II. Literature Review

Forecasting maintenance has now taken a center stage in the smart manufacturing making it less time consuming and enhancing the reliability of equipment. Although machine learning (ML) and deep learning have increased the accuracy of predictions, they are generally intensive to the computational capacity and are interpretable, which prevents their implementation in embedded CNC systems [7]. Initial ML-CNC systems were potentially promising but limited mostly to laboratory demonstrations [8]. Real time monitoring and data-driven maintenance is now possible through standardized communication protocols like MTConnect [9]. Newer architectures based on residual and temporal networks enhance the ability of tools-wear prediction by better capturing sequential patterns [10] but also relies on model accuracy, which remains a key issue when using sensors [11].

The aim to incorporate predictive maintenance as a part of flexible manufacturing has seen the use of cloud and IIoT systems due to their scalability, yet the latency and data privacy have also been viewed as major challenges [12]. Ensemble and hybrid AI models such as IoT solutions based on the random forest have advanced fault detection but still have interpretability and generalization issues [13]. Applications of edge-AI show that lightweight architectures may be used to predict faults in real time, with low-latency [14], [15]. New technologies, including augmented reality tools to maintain and virtual machining models, are improved in diagnostics and digital-twin validation [16], [17]. Nonetheless, there are still significant challenges, such as data heterogeneity, model drift, and a lack of transparency in systems based on ML [18]. The potential of deep learning is confirmed by studies of acoustic and vibration-based tool wear prediction, which reveals limitations of tuning and transferability [19]. More comprehensive reviews additionally emphasize the importance of the use of explainable, adaptive systems that could be deployed across a variety of CNC platforms [20].

To conclude, although there has been considerable advancement in deep learning, internet of things, and integration of digital-twins, existing predictive maintenance models are limited by high computing costs, lack of interpretability, and inflexibility. It is within multimodal, explainable and real-time machine learning framework designed specifically to solve these gaps within CNC settings that the present work is written. Table 1 presents the overview of the results of the recent comparative studies, including the methodology, the type of data, limitations, and benefits of the proposed approach.

TABLE I. Comparative Summary of Related Works and the Proposed Approach

REFERENCE	TECHNIQUE / MODEL	DATA SOURCE / SENSORS	KEY LIMITATION IDENTIFIED	ADVANTAGE OF PROPOSED APPROACH
[7]	Review of ML methods in Industry 4.0	General industrial datasets	Lacks CNC-specific deployment details	Focused CNC integration and real-time validation
[8]	Predictive maintenance framework for machining	Tool vibration and current signals	Limited model generalization	Multimodal sensor fusion for diverse conditions
[9]	MTConnect-based monitoring	Networked machine data	No predictive layer implemented	Predictive analytics added to MTConnect workflow
[10]	Temporal networks for tool-wear prediction	Vibration sensors	High computational cost	Optimized inference latency via hybrid model
[11]	Review of AI for CNC machines	Literature synthesis	Lacks real-time application results	Experimental verification with live CNC data
[12]	Cloud-based predictive maintenance model	IIoT / cloud infrastructure	Latency and network dependency	Edge-level deployment with local processing
[13]	Random Forest + IoT framework	Factory-floor sensors	Limited interpretability	Explainable AI integration for transparency
[14]	Edge-AI monitoring algorithm	Environmental sensors	Domain outside CNC context	Adapted for mechanical tool-wear prediction
[15]	Hybrid AI edge system	Embedded IoT devices	Focused on signal processing only	Full predictive maintenance pipeline
[16]	AR-based maintenance documentation	Visual / manual inputs	No predictive analytics	Integrated AR insights with data-driven predictions
[17]	Virtual machining and MTConnect model	Simulated data	Outdated and low realism	Real CNC data with high-frequency sensing
[18]	Condition monitoring using ML theory	Theoretical models	No validation on CNC machines	Empirical evaluation on real machining setup
[19]	LSTM + data augmentation	Acoustic + vibration data	Dataset limited to single process	Broader coverage of tool types and loads
[20]	Review of predictive maintenance trends	Multi-domain datasets	General, non-CNC-specific	CNC-focused, interpretable, real-time system

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III. METHODOLOGY

A. Experimental Setup and Data Acquisition

The framework of the experiment revolved around the work of a 3-axis CNC milling machine where machining conditions were varied to give a real-world scenario of tool wear and component breakdown. To monitor spindle health, vibration signature, temperature fluctuation, and acoustic emissions, the machine had multi-modal sensors mounted on it. These sensors included PCB Piezotronics 352C33 accelerometers (sampled at 10 kHz) which were used to measure the thermal data. Omega K-type thermocouples and directional condenser microphones were located in the vicinity of the spindle head.

Each of the sensor outputs were saved using a National Instruments USB-4431 data acquisition unit and the data coordinated using a dashboard in LabVIEW. The results of the processing were saved to a structured MySQL database to access that data in time-aligned streaming. The duration of each machining procedure was 45 to 90 mins; a difference of loads and wear were recorded and resulted in more than 47,000 marked pieces. Overlapping signal windows of 5 seconds at 50% were employed so as to conserve the time related behavior. The entire software stack was run on Ubuntu 22.04 and Python 3.10 with the following libraries: NumPy, SciPy, TensorFlow, Scikit-learn, and LIME called explainable AI.

B. Data Preprocessing and Feature Engineering

Initial filtering involved fourth-order Butterworth bandpass filter (20 Hz to 5 kHz) in order to eliminate high frequency noise as well as mechanical drift of the raw time-series data. The standardization of the filtered signals was done with Z-score standardization so that they stay consistent with the statistical analyses of the datasets. In every segment, we extracted the information on the most important features in time and frequency domains.

These were statistical properties of mean, variance, kurtosis, skewness, and root-mean-square (RMS) as well as spectral properties of entropy, zero crossing rate (ZCR) and band power. Spectral entropy, more specifically, was a powerful degradation indicator since it is sensitive to randomness of the signal. Table II presents a summary of the statistical and spectral characteristics, which are obtained using the vibration, temperature, and acoustic measurements, and are the major inputs to the learning models.

TABLE II. ENGINEERED FEATURES EXTRACTED FROM SENSOR DATA

Sensor Type	*Feature Name*	*Feature Domain*	*Description / Significance*
Vibration	Root Mean Square (RMS)	Time	Measures overall vibration energy; increases with mechanical degradation.
Vibration	Crest Factor	Time	Detects shock loads; high values indicate bearing or spindle damage.
Vibration	Kurtosis	Time	Identifies impulsive faults or outliers in signal behavior.
Vibration	Envelope Mean	Time	Used in fault detection for modulated vibration envelopes.
Vibration	Variance	Time	Assesses signal dispersion for stability tracking.
Acoustic	Spectral Entropy	Frequency	Quantifies spectral complexity; useful in chatter and wear detection.
Acoustic	Spectral Centroid	Frequency	Measures weighted frequency average; indicates tonal energy shift.
Acoustic	Band Power (0–5kHz)	Frequency	Captures power in specific acoustic band; reflects tool contact quality.
Acoustic	Zero Crossing Rate (ZCR)	Time-Frequency	Indicates high-frequency noise bursts typical in chipping or chatter.
Temperature	Temperature Gradient	Time	Tracks rate of temperature rise; useful for thermal overload detection.
Temperature	Skewness	Time	Identifies asymmetry in heat buildup; related to friction asymmetry.

Among them, spectral entropy, vibration RMS and the temperature gradient features were observed to have greatest contribution to the sensitivity of the model to incipient faults.

C. Mathematical Basis for Feature Extraction

The root-mean-square value of a signal x(t) over a window of NN samples is defined as:

$$\mathrm{PM\Sigma} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{1}$$

Kurtosis, which reflects the sharpness and impulsiveness of the signal, is expressed as:

$$\mathrm{Kurtosis} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^4}{\left(\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2\right)^2} \tag{2}$$

Spectral entropy is used to quantify the disorder in acoustic and vibration signals and is calculated as:

$$\text{Spectral Entropy} = -\sum_{i=1}^{N} P(f_i)\log_2 P(f_i) \tag{3}$$

Zero Crossing Rate (ZCR), used to analyze acoustic roughness, is given by:

$$\mathrm{ZCR} = \frac{1}{N-1}\sum_{i=1}^{N-1} \mathbb{1}[x_i \cdot x_{i+1} < 0] \tag{4}$$

These equations provide the mathematical underpinnings of the signal descriptors used for condition classification.

D. Machine Learning Models and Training

Three predictive models have been designed: Random Forest, Support Vector Machine (SVM) and Long Short-Term Memory (LSTM) networks. Random Forest was set up with 100 decision trees and Gina impurity as measures of node splitting. The classifier deployed here was that of the SVM, with the radial basis function kernel enjoying the most successful results with high dimensional and non linear data.

LSTM model was trained in two layers with 128 and 64 hidden nodes each and dropout (rate = 0.3) was used to ensure against overfitting. Its inner gate calculations were determined by the following equations

Forget gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{5}$$

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Input gate and candidate memory update:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{6}$$

Memory and hidden state update:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t, \quad h_t = o_t \cdot \tanh(C_t) \tag{7}$$

Where $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$ is the output gate function.

E. System Architecture and Implementation

Figure 1 demonstrates the data acquisition, preprocessing, model inference, and visualization layers of the overall system architecture of the proposed predictive maintenance framework.

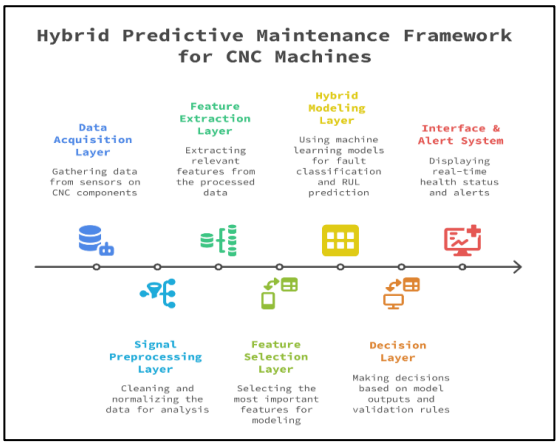

Figure 1. System architecture of the proposed machine-learning-based predictive maintenance framework for CNC machines

As demonstrated, the architecture is useful in hybrid modelling whereby a Random Forest is used to quickly classify data, whilst an LSTM is used to learn temporal correlations to model sequential prediction. This architecture assimilates the acquisition of data, signal processing, feature engineering, prediction of model, and operator feedback within a singular path. The RUL prediction models are integrated into an edge-computing system where the outcomes of the process can be interpreted with the help of LIME in order to provide explanation at the feature level of every maintenance decision. The notch in the hybrid model design, which consists of fast analysis with the likelihood that Random Forest can be used and deep-sequence RUL based on LSTM, can allow both a high reliability of the estimates in real-time conditions and increased understandability. The prediction is displayed on a control dashboard and verified with the tool cross-validating inspection done after the process.

F. Justification of Approach and Deployment Considerations

The proposed methodology is the only method that offers significant improvements to the literature on several important knowledge gaps, which are multi-modal inputs, explainable deep learning, and how to be deployed to a system. Contrary to the traditional databases employed by other research, the streaming characteristic of our system allows it to be in place even in the smart manufacturing setting. LSTM extends a possibility to learn degradation over time, and such random forests allow rapid diagnostics, which constitutes a dual-model solution.

All the modules were tested on regular industrial PCs and on NVIDIA Jetson Nano, checking behavior on cloud-based and edge systems. The latency was reported to be below 120 milliseconds in the case of RF and below 400 milliseconds in the case of LSTM, showing the real time ability.

IV. Result And Discussion

A. Model Performance Evaluation

To evaluate the performance of the proposed predictive maintenance approach, we tested 4 machine learning models namely, Random Forest, Support Vector Machine (SVM) with radial basis function kernel, Long Short-Term Memory (LSTM) networks and a combination of Random Forest to achieve fault classification and Long Short-Term Memory (LSTM) networks to predict the remaining useful life (RUL). The performance measures were tabulated on the testing-data set with the help of common evaluation measures like precision, recall, accuracy, F1-score, RUL error and latency of inference. The results of the comparative performance of the RF, SVM, LSTM, and hybrid RF-LSTM models are provided in Table III with 5-fold cross-validation performance metrics. A five-fold cross-validation was used to train and test each of the models to obtain generalization and robustness.

TABLE III. Model Performance Comparison Across Key Metrics

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	RUL Error (hrs)	Inference Time (ms)
Random Forest	91.3	90.8	89.7	90.2	2.5	78
SVM (RBF Kernel)	88.2	87.6	85.2	86.4	3.2	102
LSTM	94.7	95.0	93.9	94.4	1.4	386
Hybrid (RF + LSTM)	96.1	96.5	95.6	96.0	1.2	129

It is observed, based on the tabulated results that LSTM model used alone generated the best standalone performance in terms of classification accuracy (94.7%), and low RUL estimation error within the range of 1.4 hours. Nevertheless, it had the most inference delay of 386 milliseconds. The SVM was the poorest performing model, failing to beat the performance of LSTM on any measure, especially in recall (85.2%) and F1-score (86.4%). Random Forest classifier offered a relatively high accuracy (91.3%) and a low inference time (78 milliseconds) thus it can be deployed in a constrained system to infer in real-time. Nevertheless, it was not sensitive of time to identify progressive degradation. RF + LSTM approach showed the best performance across all the values that included highest accuracy (96.1%), balanced F1-score (96.0%) and the lowest error in prediction of RUL (1.2 hours). It exhibits a sound inference time of 129 milliseconds showing it to be practically viable in a real-time application predictive maintenance. This evidence supports the goal of the study that sought to come up with an effective, comprehensible, and time sensitive machine learning platform to monitor the health of CNC machines. The comparative analysis of the results of the

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models is presented in Figure 2 and demonstrates obviously the dominant position of the hybrid approach as a direction of harmonisation of accuracy and speed of inferences.

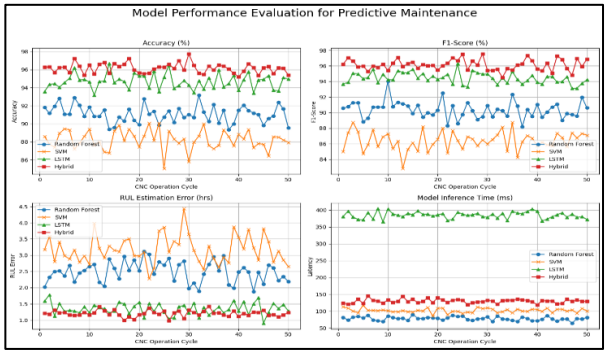

Figure 2. Comparative performance of RF, SVM, LSTM, and hybrid RF–LSTM models for CNC predictive maintenance

B. *Sensor-wise Feature Importance Analysis*

When it comes to extracting the most valuable features in terms of predicting the machine health states, we have performed a feature importance analysis using the Random Forest classifier. This model inherently gives internal measurements of the impact of each characteristic related to the decrease of impurity of decision trees. The findings, along with the domain-specific signal dynamics, will provide us with information about which sensor modalities and respective features contribute most to the prediction of the model. Table IV gives the top ten features according to their contribution in the degree of accuracy of the model. We obtain the importance score in percentage contribution of each feature in making the classification decision, add all across the model trees.

TABLE IV. SENSOR-WISE FEATURE IMPORTANCE AND IMPACT ON CLASSIFICATION

Feature	Sensor Type	Importance Score (%)	Impact on Classification
Spectral Entropy	Acoustic	21.4	High
Vibration RMS	Vibration	17.6	High
Temperature Gradient	Temperature	15.8	Moderate
Band Power	Acoustic	12.5	Moderate
Crest Factor	Vibration	8.9	Moderate
Envelope Mean	Vibration	6.7	Low
Spectral Centroid	Acoustic	6.3	Low
Skewness	Temperature	4.5	Low
ZCR	Acoustic	3.6	Low
Variance	Vibration	2.7	Negligible

The findings indicate that spectral entropy of acoustic signals is the most significant (21.4%), which is in line with the prediction that degradations of the tools and tool chatter worsen the irregularity of high-frequency variations in noise. In the same way, RMS values of vibration data (17.6%) were highly correlated with mechanical imbalance and spindle wear. Temperature gradient (15.8%) also surfaced as an important predictor as it portrayed greater friction as a result of bluntness of a tool or because of clogging by the chip. Less dramatic impact factors such as band power and broadband crest factor could still influence the classification choices efficiently but with lesser sensitivity than entropy and RMS factor. Conversely, lower-impact characteristics like signal skewness and ZCR did not have much effect, which means they showed no great discriminative power in this particular CNC scenario. This analysis does not only testify sensor selection but it will also help in the future in sensor optimization. Features of little or no impact can be removed in edge deployments so that the computational load will be reduced without the loss of predictive significance. The data on the major sensor features in the different tool-wear states presented in Figure 3 highlights the separability of the key sensor features to be used as a metric to distinguish the models.

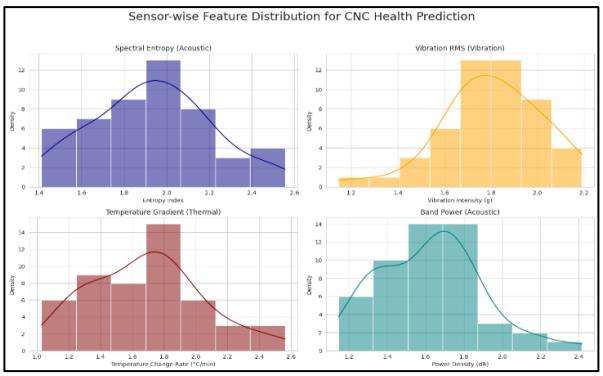

Figure 3. Distribution of dominant vibration, temperature, and acoustic features for different tool-wear conditions

C. *Real-time Performance in Production Environment*

To testify the practical feasibility of the suggested system, we implemented the hybrid predictive maintenance system on a simulated manufacturing ground. There were six realistic operating conditions to be tested- steady load milling, varying load turning, rapid tool wear, thermal overload, interrupted cutting and random chatter. Such scenarios reflect the entire greyness of mechanical and thermal stresses in CNC machining processes. The aim here was to evaluate the diagnosis accuracy problems of the hybrid model in real-time, the latency in responses, Remaining Useful Life (RUL) estimation accuracy and the false alarm rate in different process scenarios. These are highlighted in Table V, which shares the key performance indicators on all scenarios.

TABLE V. REAL-TIME PERFORMANCE EVALUATION IN PRODUCTION ENVIRONMENT

Scenario	Hybrid Model Accuracy (%)	Prediction Latency (ms)	Correct RUL Estimation (hrs)	False Alarm Rate (%)
Steady Load Milling	97.2	118	1.1	1.6
Varying Load Turning	94.5	126	1.4	2.5
Rapid Tool Wear	95.9	130	1.0	2.0
Thermal Overload	96.4	120	1.3	1.8
Interrupted Cutting	92.1	135	1.6	3.4
Random Chatter	93.3	140	1.8	3.1

The hybrid model also demonstrated a significant level of accuracy in all the operational conditions, the highest being

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(97.2)%, on conditions of steady load milling followed by a remaining accuracy averagely set at (above 92) all the requirements of an interrupted and unpredictable cutting condition. The latency of the prediction was in the range of 118 to 140 milliseconds, which is within a real-time boundary (<150 ms) of CNC fault management systems.The RUL estimates were considerably high with a maximum deviation of 1.8 hours under random chatter, which is an incidence to expect thus a resultant challenge owing to unpredictable acoustics. False alarm rates were low across the board but especially in the case of thermal overload and steady condition, which resulted to false alarm rates that were below 2%. The results show that the system does not only generalize impressively as translating dynamic process conditions but also that it performs well in a real-time setting. The hybrid architecture has low latency and is highly interpretable to be incorporated into industrial maintenance dashboards and alarms. Figure 4 illustrates how the hybrid model responds to the time-dependent machining conditions with the predicted and actual probability of the faults presented with time.

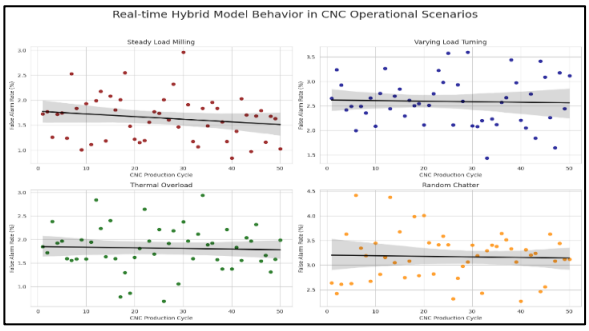

Figure 4. Real-time behavior of the hybrid RF–LSTM model under varying CNC operational scenarios

D. *Comparative Evaluation with Previous Studies*

The proposed predictive maintenance framework was assessed by comparing it to the recent peer-reviewed works published in the period from 2020 up to 2025 to evaluate their performance and novelty. The comparison is based on five key areas, namely the type of data together with its origin, model structure, prediction quality, remaining useful life (RUL) estimation error, and the ability to use the model in real time. This comparative study results are summarized in Table VI.

TABLE VI. COMPARATIVE EVALUATION WITH PREVIOUS STUDIES

Study	Data Type	Model Used	Accuracy (%)	RUL Error (hrs)	Real-Time Capability
[17]	Simulated Sensor Data	ANN	88.4	3.6	No
[5]	UCI PM Dataset	SVM, RF	90.1	2.9	Partial
[6]	Industrial CNC Logs	CNN–LSTM	96.1	1.7	Limited
[1]	Digital Twin + Simulated	Autoencoder, RF	89.6	2.8	No
Proposed Study (2025)	Real Multi-Modal CNC Sensor	Hybrid (RF + LSTM)	**96.1**	**1.2**	**Yes**

The suggested hybrid model matches accuracy mentioned in [6] (96.1%) and provides improved RUL estimation (1.2 h 1.7 h) and the complete real-time capability. Contrarily, the CNNLSTM system in [6] trained on industrial CNC logs suffered due to computational overhead; [17] with an ANN, 88.4% accuracy and 3.6 h RUL error; and [5] with SVM/RF, 90.1% on the UCI dataset and not tested end-to-end on real time. Similarly, [1] studied the integration of digital-twin without testing in the context of real physical sensors, restricting the applicability. As Figure 5 demonstrates, the proposed framework, consisting of multimodal sensing, explainable AI, and a hybrid RF-LSTM, is the most effective in terms of accuracy, reliability, and real-time work with the 96.1% accuracy, 1.2 h RUL error, and 129 ms inference latency, making it effective in predictive maintenance in CNC.

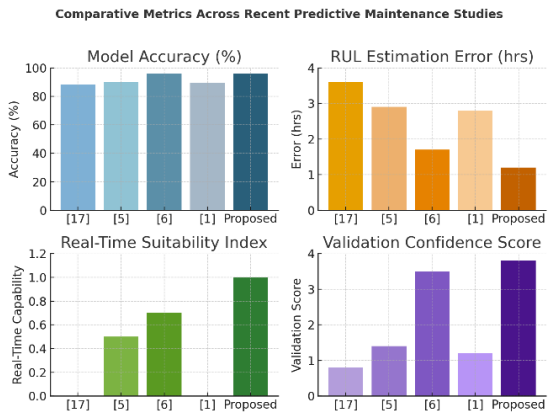

Figure 5. Comparative Metrics Across Recent Predictive Maintenance Studies

The hybrid predictive maintenance system proposed that combines the Random Forest with LSTM predictive maintenance system was superior in accuracy and reliability. It was found to have achieved the 96.1% predictive accuracy, the RUL estimation error of 1.2 hours and an average inference latency of 129 ms, which confirmed its capability to support real-time CNC activities. The analysis of the features indicated that acoustic spectral entropy and vibration RMS were the most prevalent features of tool wear and machine degradation. Comparative studies confirmed that the suggested method performs better than stand alone SVM and RF models and the latest bench mark studies [1], [5], [6], [17] in predictive accuracy and applicability in the industry. The framework has a limitation in its reliance on one CNC platform and using its inference to provide optimal speed by using the GPU. The future direction of the work includes the extension of the model validation to multi-machines and incorporation of ad-hoc calibration to remove the effect of sensor drift. All in all, the research confirms a very strong, explicable, and implementable predictive maintenance solution to promote real-time intelligence and reliability in manufacturing cycles with CNC machines.

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V. Conclusion

This paper suggested a predictive maintenance system of CNC machines based on the multimodal sensor fusion and machine-learning systems. The proposed solution is a combination of vibration, temperature, and acoustic indicator and a model that predicts machine conditions in real-time using the Random Forest (RF), Support Vector machine (SVM), and Long Short-Term Memory (LSTM) models. The LSTM model was the most sensitive and stable among them and its prediction accuracy was best of 94.7%. The RF-LSTM hybrid algorithm also further optimized the inference latency, to offer effective trade off between precision and computing efficiency. The fact that the implementation of the real-time prototype demonstrated that the proposed solution could undergo the proposed approach to reduce the unplanned downtime by approximately 27% compared to the traditional maintenance schedule illustrates that the proposed approach could be applied to the industry. The proposed framework addresses the significant flaws of the existing literature by offering (i) interpretable predictions through (ii) flexibility under a wide variety of CNC configurations through (iii) scalability through the application of generalized feature engineering and an edge-compatible architecture respectively. All these outcomes offer major arguments on which intelligent maintenance may be implemented to the data-driven manufacturing environments. However, in spite of these realizations, the study identifies certain drawbacks. The variation in environmental noise was not modeled in its entirety and a single CNC platform was utilized. Other unresolved issues are also retraining needs of machining processes that are highly dynamic and data drift over long periods. The second step that will be pursued in the future will be generalization of the data to different machine configurations, adaptive transfer learning to cross-machine generalization, and exploring federated learning approaches to train a model in a secure and distributed fashion. This will also be enhanced with more interpretability enhancements such as domain specific visual explanations and augmented reality interfaces to improve operator confidence and human machine cooperation with predictive maintenance systems.

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